

LUDINGTON HARBOR, MI, USA

WMO: 997261

Lat: 43.947N

Lon: 86.441W

Elev: 179

StdP: 99.19

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-15.3	-13.2	N/A	N/A	N/A	N/A	N/A	N/A	11.8	-1.7	10.7	0.0	3.9	300	0.521

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.5	27.0	N/A	25.9	N/A	25.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	4.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
10.6	9.0	8.0		-18.8	29.7	2.7	2.7	-20.7	31.6	-22.3	33.2	-23.8	34.8	-25.8	36.8	37.8
				N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
				DB												
				WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	-4.0	-4.1	0.4	6.8	12.4	17.8	20.0	19.8	16.3	10.2	4.3	-0.8	(6)	
(7)		DBStd	9.70	5.10	4.69	5.51	4.41	4.42	3.42	3.07	2.70	3.79	4.37	4.51	4.45	(7)	
(8)		HDD10.0	1837	435	394	303	117	24	1	0	0	1	50	179	333	(8)	
(9)		HDD18.3	3853	693	627	557	346	193	51	18	15	84	254	421	592	(9)	
(10)		CDD10.0	1228	0	0	5	21	98	233	312	305	189	58	8	0	(10)	
(11)		CDD18.3	203	0	0	0	1	8	34	72	62	23	3	0	0	(11)	
(12)		CDH23.3	676	0	0	0	1	38	132	258	173	65	9	0	0	(12)	
(13)		CDH26.7	46	0	0	0	0	3	11	19	9	3	0	0	0	(13)	
(14)	Wind	WSAvg	3.4	3.9	3.7	3.4	3.6	3.4	2.9	2.8	2.7	2.9	3.5	4.0	3.9	(14)	
(15)	Precipitation	PrecAvg	848	51	37	52	74	66	77	54	101	103	86	80	68	(15)	
(16)		PrecMax	1075	115	91	180	118	139	211	108	238	272	196	186	163	(16)	
(17)		PrecMin	616	11	6	8	25	10	5	16	8	6	26	16	23	(17)	
(18)		PrecStd	118	29	23	39	26	35	45	27	57	62	38	39	35	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	8.9	20.4	22.6	27.1	28.3	29.0	27.8	27.1	24.8	17.6	13.3	(19)	
(20)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)
(21)		2%	DB	6.4	5.5	15.7	18.8	24.3	26.2	26.9	26.3	25.1	22.0	15.1	8.5	(21)	
(22)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)
(23)		5%	DB	4.7	3.5	11.2	15.9	21.7	24.9	25.8	25.1	23.8	19.2	12.8	6.8	(23)	
(24)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)
(25)		10%	DB	3.2	2.1	8.0	13.5	19.3	23.5	24.9	24.2	22.4	16.9	11.0	5.3	(25)	
(26)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)	
(28)			MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)
(29)		2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)
(30)			MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)
(31)		5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)
(32)			MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)
(33)		10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)
(34)			MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)
(35)	Mean Daily Temperature Range	MDBR		5.2	5.6	7.1	7.7	8.2	7.7	6.5	6.5	7.6	6.8	5.8	4.7	(35)	
(36)		5% DB	MADB	7.0	6.7	9.7	11.1	10.7	9.2	7.0	6.3	8.1	8.8	8.3	6.8	(36)	
(37)			MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)
(38)		5% WB	MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)
(39)			MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.293	0.312	0.335	0.367	0.397	0.409	0.411	0.402	0.384	0.349	0.317	0.289	(40)	
(41)		taud		2.438	2.372	2.394	2.334	2.316	2.325	2.329	2.351	2.382	2.468	2.497	2.487	(41)	
(42)		Ebn at Noon		838	879	903	897	875	863	856	852	839	828	804	810	(42)	
(43)		Edh at Noon		73	93	104	120	126	125	123	117	105	84	69	64	(43)	
(44)	All-Sky Solar Radiation	RadAvg		1.10	1.97	3.53	4.74	5.70	6.27	6.38	5.41	4.24	2.56	1.43	0.88	(44)	
(45)		RadStd		0.13	0.23	0.37	0.44	0.49	0.42	0.29	0.34	0.36	0.30	0.21	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page